

Coupling & Resonance L/C Design Values

Resonant electrophysiology — TEST1 (verified) and X10 10-channel plan · ngspice-42 simulations, closed-loop Xschem verification · Aug 18, 2026 · companion file: LC_design_values.xlsx (live formulas)

1 · TEST1 — verified design point (both channels active)

Element / channel	L/branch (nH)	Rser (Ω)	C res. (pF)	C coup. (fF)	f0 sim (MHz)	Depth (dB)	BW–3dB (kHz)	QL	Itail (mA)	Margin (%)
Channel A (active line)	50	0.5	101	280	55.890	35.9	23.2	2413	4.07509	1.55
Channel B (active line)	50	0.5	40.362	155	100.002	39.7	13.9	7217	0.65260	0.90
act1 (isolated ref.)	50	0.5	101	400	49.94	0.14	—	—	0.130	—
Passive tank 1	100	1.0	101	400	49.98	0.02	—	—	—	—
Passive tank 2	100	1.0	101	70 000	38.44	28.1	—	—	—	—

Channel B tank retuned 60 → 40.362 pF to land exactly 100.00 MHz (naive calculation gives 42.2 pF; the shared act– return through L6 || L4 = 25 nH shifts every mode up — e.g. channel A: naive 50.0 → simulated 55.89 MHz). Spec: BW ≤ 30 kHz per channel (QL ≥ 1667 @ 50 MHz) — both channels PASS. Sensitivity: ±0.25% Itail keeps depth ≥ 17 dB; +1% on Itail_B → oscillation (bias feedback/AGC required in hardware).

2 · X10 — 10-channel plan, 50–200 MHz

Ch	f target (MHz)	C tank yours (pF, L=30 nH)	Populated in .asc	Q0 yours	C tank rec. (pF, L=100 nH)	cp rec. (fF)	QL spec (f/30 kHz)	Itail est. (mA)	P est. (mW)
1	50.000	337.737	c10	6.3	101.01	312.1	1667	5.793	8.69
2	66.667	189.977	—	8.4	56.75	234.1	2222	2.343	3.51
3	83.333	121.585	—	10.5	36.28	187.2	2778	1.157	1.74
4	100.000	84.434	c6	12.6	25.17	156.0	3333	0.653	0.98
5	116.667	62.033	—	14.7	18.45	133.7	3889	0.403	0.60
6	133.333	47.494	—	16.8	14.09	117.0	4444	0.265	0.40
7	150.000	37.526	c8	18.8	11.15	104.0	5000	0.182	0.27
8	166.667	30.396	—	20.9	9.03	93.6	5556	0.130	0.19
9	183.333	25.121	—	23.0	7.45	85.1	6111	0.095	0.14
10	200.000	21.108	c1	25.1	6.26	78.0	6667	0.074	0.11
							TOTAL:	11.093	16.64

As-is verdict: with your current X10 values (L=30 nH, RS=1.5 Ω , Kp=200 $\mu\text{A}/\text{V}^2$, Itail=170–190 μA) the line shows NO measurable notches (<0.03 dB) — Q0 = $\omega\text{L}/\text{RS}$ is only 6–25 and the pairs barely compensate. Note the populated-channel comment labels in the .asc are stale (the block labeled “50MHZ” carries c1 = 21.1 pF → 200 MHz).

3 · Design rules (anchored on the two verified points)

- Notch (series) resonance through the coupling cap: $f_0 = 1 / (2\pi \cdot \sqrt{L_{\text{loop}} \cdot (C_{\text{tank}} + c_p)})$
- Tank quality factor: $Q_0 = 2\pi \cdot f \cdot L_{\text{loop}} / R_{S,\text{loop}}$ — spec: $QL = f_0 / BW \geq f_0 / 30 \text{ kHz}$
- Coupling rule: $c_p(f) = 1 / (2\pi \cdot f \cdot Z_{cp})$ with $Z_{cp} \approx 10.2 \text{ k}\Omega$ (check: 280 fF @ 55.89 MHz, 155 fF @ 100.00 MHz — both verified)
- Bias estimate: $I_{\text{tail}}(f) \approx 0.6526 \text{ mA} \cdot (100 \text{ MHz} / f)^{3.15}$ (two-point empirical fit — verify by co-tuning in simulation)
- Q enhancement: $Q_{\text{enh}} = Q_0 / (1 - g_m R_p / 2)$, oscillation at $g_m R_p / 2 = 1$; $g_m = \sqrt{(2 \cdot K_p \cdot (W/L) \cdot I_{\text{tail}} / 2)}$

Method checklist (validated on TEST1)

1. Full-span AC to locate the real modes (shared act– return shifts them up from the naive f_0).
2. Re-measure the oscillation threshold AT each c_p (coupling loss raises it); park 0.5–3% below.
3. Map ($c_p \times$ bias margin) $\rightarrow$ pick maximum depth subject to $BW \leq 30 \text{ kHz}$ (depth ridge at critical compensation).
4. Co-tune all channels jointly — thresholds interact through the shared return ($\approx 9\%$ shift seen in TEST1).
5. Trim each tank C for exact channel frequency (TEST1: C7 60 $\rightarrow$ 40.362 pF for 100.002 MHz).
6. Sensitivity audit: $\pm 0.25\%$ $I_{\text{tail}} \Rightarrow$ depth $\geq 17 \text{ dB}$; +1% on channel B $\Rightarrow$ oscillates. Plan an AGC/bias loop.

Sources: CROSS COUPLED1_100MHz_design.asc (TEST1) · CROSS COUPLEDX10.asc (.param C1...C10) · ngspice-42, 2026-08-18 · Xschem closed-loop verification in simulaciones/ · companion Excel with live formulas: LC_design_values.xlsx